

Grothendieck group work: commentary

NSFLEAPS/_07_RP/RP_Nat_ext_double.lean

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These are the explanatory comments added alongside the Grothendieck group material of the file: Theorem 6.1 (the group `groGroup` and the map `groGroupHom`), Theorem 6.3 (the action of `groGroup S` on `X`), and Corollary 6.4. Pre-existing comments are omitted, as are the proofs themselves; each note is labelled with the declaration it precedes and its line number in the file.

attribute [reducible] groGroup line 104

`groGroup` is made reducible so that `groGroup S` and `Algebra.GrothendieckGroup (WithOne S)` are interchangeable during type class synthesis. This only adds an attribute; the definition above is unchanged.

def groGroupOf line 137

`Algebra.GrothendieckGroup.of`, viewed as a map from `WithOne S` into `groGroup S`

lemma groGroupOfMul line 143

`groGroupOf` is a monoid homomorphism

lemma groGroupHomEqOf line 149

`groGroupHom` is the restriction of `groGroupOf` to `S`

lemma groGroupOfEqMk line 154

`groGroupOf` written using `Localization.mk`

lemma withOneCases line 160

Case analysis on `WithOne S`

lemma groGroupIsQuotient line 214

Every element of `groGroup S` is a quotient of two elements of `WithOne S`

lemma withOneMulCoe line 232

Multiplying an element of `WithOne S` by an element of `S` lands in the image of `S`

lemma imageOfSemiInGGMulSelf line 240

The image of `S` in `groGroup S` is closed under right multiplication by `groGroupHom`

lemma existsShiftIntoImageOfSemiInGG line 248

Every single element of `groGroup S` can be shifted into the image of `S`. This is the one-element case of `imageOfSIsThickInGroGroup`.

def actionHomeo line 305

Theorem 6.3: the action of the Grothendieck group

When `S` acts by homeomorphisms, the maps `f_s` all commute, so the subgroup of $X \simeq_t X$ that they generate is commutative. The universal property of the Grothendieck group then extends the action of `S` to an action of `groGroup S`, which is what the paper constructs by hand.

def actionHomeo line 312

When `S` acts by homeomorphisms, the homeomorphism of `X` given by `s`

lemma actionHomeoMul line 326

`actionHomeo` turns multiplication in `S` into composition of homeomorphisms

def actionSubgroup line 334

The subgroup of $X \simeq_t X$ generated by the action of `S`

lemma actionSubgroupGensComm line 342

The generators of `actionSubgroup` commute, because `S` is commutative

instance actionSubgroupCommGroup line 352

Hence `actionSubgroup` is a commutative group

def actionMulHom line 362

`actionHomeo`, viewed as a semigroup homomorphism into `actionSubgroup`

def groGroupHomeoHom line 373

The extension of the action of `S` to `groGroup S`, via the universal property of the Grothendieck group

lemma groGroupHomeoHomApplyHom line 382

The extended action restricts to the original action along `groGroupHom`

lemma dynamicalSystemEq line 410

Two dynamical systems with the same action map are equal

lemma groGroupDynamicalSystemApplyHom line 434

The extended action restricted along `groGroupHom` is the original action

lemma orbitSubsetGroGroupOrbit line 444

The `S`-orbit of a point is contained in its `groGroup S`-orbit

lemma groGroupSystemMinimalOfMinimal line 453

If the `S`-action is minimal then so is the extended `groGroup S`-action

lemma groGroupActionCommutates line 465

The actions of `S` and of `groGroup S` on `X` commute

Unconditionally, RP for the original action is contained in RP for the extended action: the `groGroup` `S`-orbit of a set contains its `S`-orbit.

If `S` acts by homeomorphisms, then RP for the original action is the same as RP for the extended action, `groGroupDynamicalSystem`.

DGG/CLAUDE: **the hypothesis `hMin` was added.** This statement is Corollary 6.4 of the paper, whose proof reads “Apply Theorem D, noting that `S` and `Gr(S)` are commutative semigroups whose actions commute and are minimal”. Theorem D is `forTwoMinCommActionsRPsAreSame`, which requires *both* actions to be minimal, so minimality of the `S`-system is genuinely needed and is indeed assumed in Corollary 6.4. Without it only the inclusion `RPSubsetExtendedActionRP` above is available: the reverse inclusion would require replacing an arbitrary $g = i(t)i(s)^{-1} \in \text{Gr}(S)$ by an element of `S`, which is exactly what minimality buys via Theorem D.